

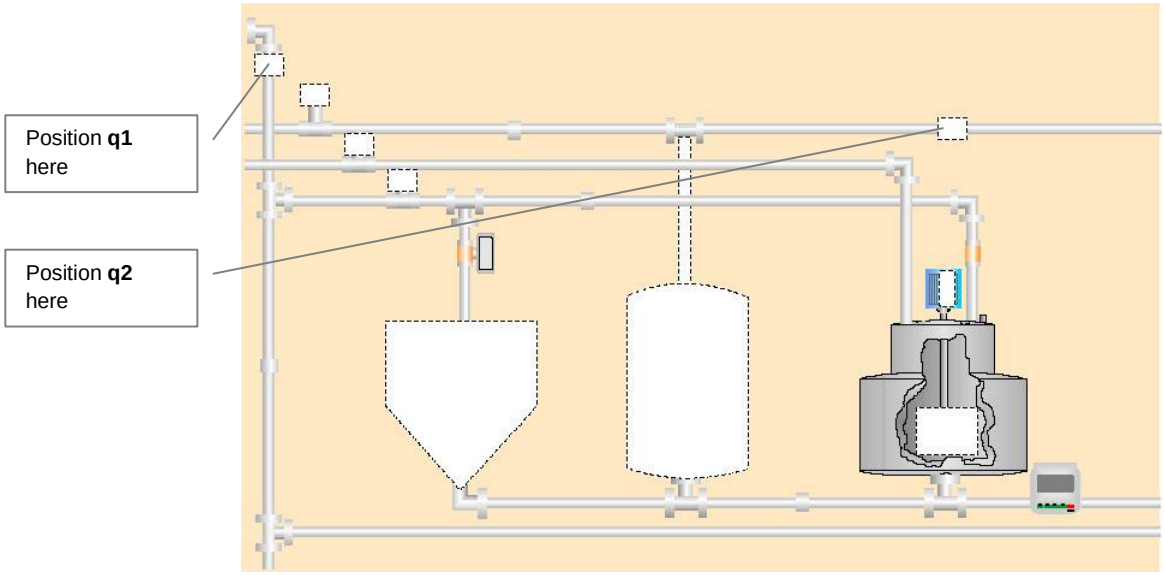
Exercise: Creating Process Components (IGSS Objects) on the Process Visualization Diagram

Purpose Learn how to create new objects for the mimic and how to use the copy/paste functions to replicate objects of identical nature.
(Note: In the next exercise, we'll learn to work with templates, which is another method for replicating objects with similar characteristics.)

Duration Max. 45 minutes.

Task 1: Create the first flow gauge for the mimic.

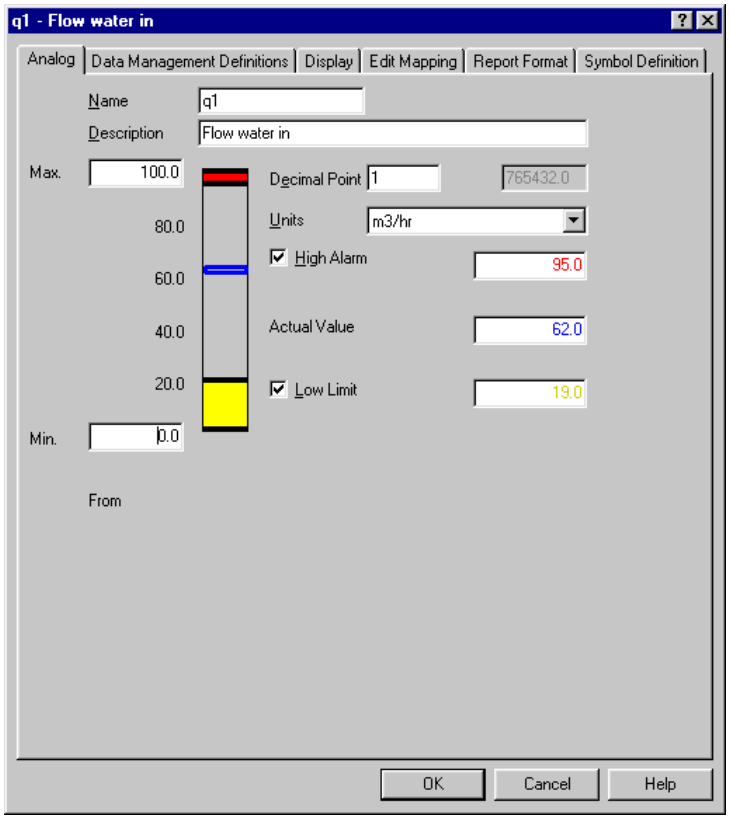
We will create two analog objects and name them **q1** and **q2** which measure water flow.



Step	Action
1.	Right-click the mouse at the position where **q1** is to be placed (see above) and in the popup menu, choose **New – Analog Elements** and the **Object Browser** appears.
2.	*TIP:* In the tree view to the left, click on **Workshop**, then on **Analog** and then select the check box **Advanced** at the bottom left to unfold the rest of the dialog box.



3. *TIP:* At the bottom right in the **Name** field, key in **q1** and click **Create** and the object properties dialog box appears as in the picture below.



4. *TIP:* Go first to the **Edit Mapping** tab and under **Atoms *remove*** the checks from **High Limit**, **Set Point,** and **Low Alarm** and leave the rest selected.

5. *TIP:* Key in or select the settings in the table below to complete the creation of the **q1** process component.

Tab Name	***Parameter***	***Input parameter(s)***
Analog	**Description**	`Flow water in`
	Max./Min (range)	Max. = `100` and Min. = `0`
	Decimal Point (number of decimals shown.)	`1`
	Units	(Leave blank until you come to optional task at the end of this exercise)
	High Alarm (select the check box)	`95.0`
	Actual Value	`62.0` (initial value)



	Low Limit **(select the check box)**	19.0

Now go to the tab called **Edit Mapping**.

Tab Name	*Parameter*	*Input parameter(s)*
Edit Mapping	**Atom** **Alarm Detatils** **PLC Node for object q1**	Set the **I/O mode:** to **local** for **High Alarm,** to **in** for **Actual Value** and to **local** for **Low Limit** Put the cursor first on **High Alarm** and click **New** to enter the **Edit Alarm Description** dialog box and create the alarm text High alarm on flow**,** :**Priority** 10, **Instructions**: Reduce inlet! (No other settings or changes are needed.) Do the same for **Low Limit,** with **Alarm text** Low limit reached**,** **Priority** 5, **Instructions**: Open inlet!. Under **Atom**, place the cursor again on **Actual Value** and then make sure the **Driver:** field shows the PLC driver we chose in **System Configuration** and that **Node:** is set to **0**. Under PLC Address for Actual Value atom key in the following: **Data Group:** 6 **Word Offset:** 0 **Bit Offset:** 1
Data Management Definition	**Scan interval**	2000 milliseconds.
	Logging	**All changes**
	Base interval	**10 min**.
	Data Reduction	**Average**
	Transfer to History	**Reduced value**
Display	**Show** **Name and State/Value**	✓ **Enable**



Symbol Definition	In the **Choose symbol** group **in Symbol Table** choose **Analog elements**…	and choose the symbol below  .

Now you're finished creating the first flow gauge, **q1**. Go on to create the next one, which you name **q2**.

Task 2: Create the second flow gauge for the mimic.

Now we'll create the **q2** gauge by copying it from **q1**.

Step	Action
1.	Right click on **q1** and choose **Copy** from the drop down menu.
2.	Right click again and choose **Paste** and then **As new object** and the original **q1** is replicated but with a new name. (Why is this?)
3.	Drag the new object to its correct position on the right side of the mimic (see the graphic above) and then rename it to **q2** and change the PLC address (**Edit Mapping** tab) of the atom **Actual Value**. (Why do this?)

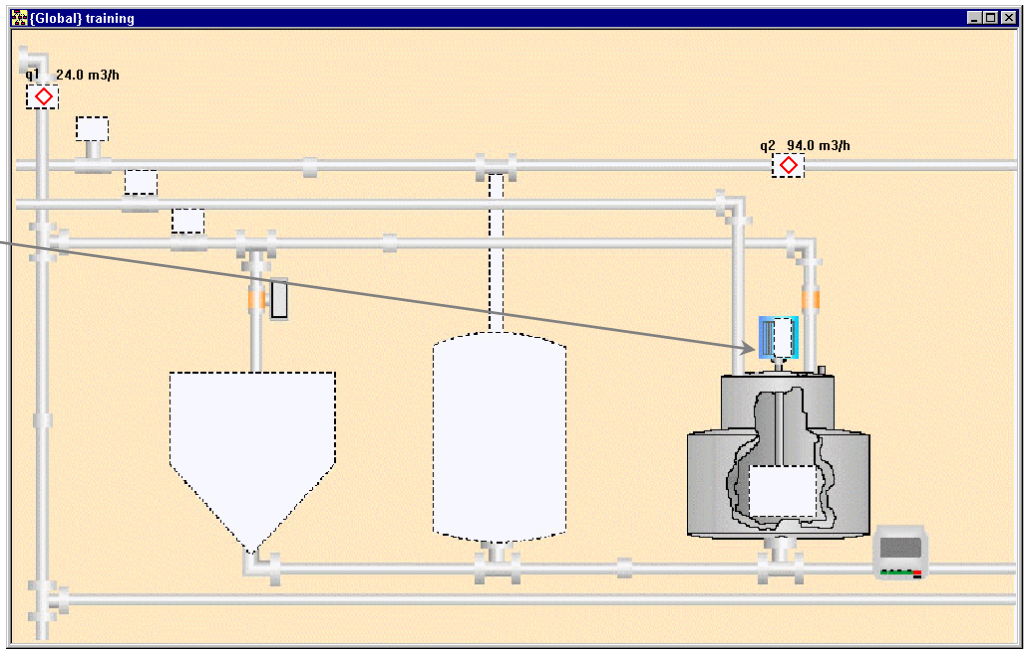
This completes creating the two flow gauges.

Task 3: Create a counter object

Now we'll create another type of IGSS object called a **Counter** object. This object, called **C1**, will be used to count the operational hours of a mixer motor, which we'll create at a later time. For now, we'll create the object and place it as shown on the diagram below.



Position **C1** here



Step	Action
1.	We will use a new graphical element to represent this object. On the menu go **Objects → Rectangular Field** and in the tree view of the **Browser** select **Workshop** then click on **Counter**.
2.	Key in the name `C1` and click **Create**. When the object properties dialog box appears, key in in the **Desc.** field `operational hours for mixer`.
3	Now follow carefully the instructions below for the remainder of the settings for the **C1** object.

Tab Name	***Setting***	***Input parameter(s)***
COUNTER	**Units**	(Leave blank until optional task at the end of exercise)
	Current count **Limit (check)** **Preset value** **Maximum value**	Current count = `100` (initial value), Limit = `150`, Preset = `15` and Max. = `200`
DATA MANAGEMENT DEF.	**Scan interval**	**5000** milliseconds
	Logging	**>5 %**
	Base interval	**10** min.



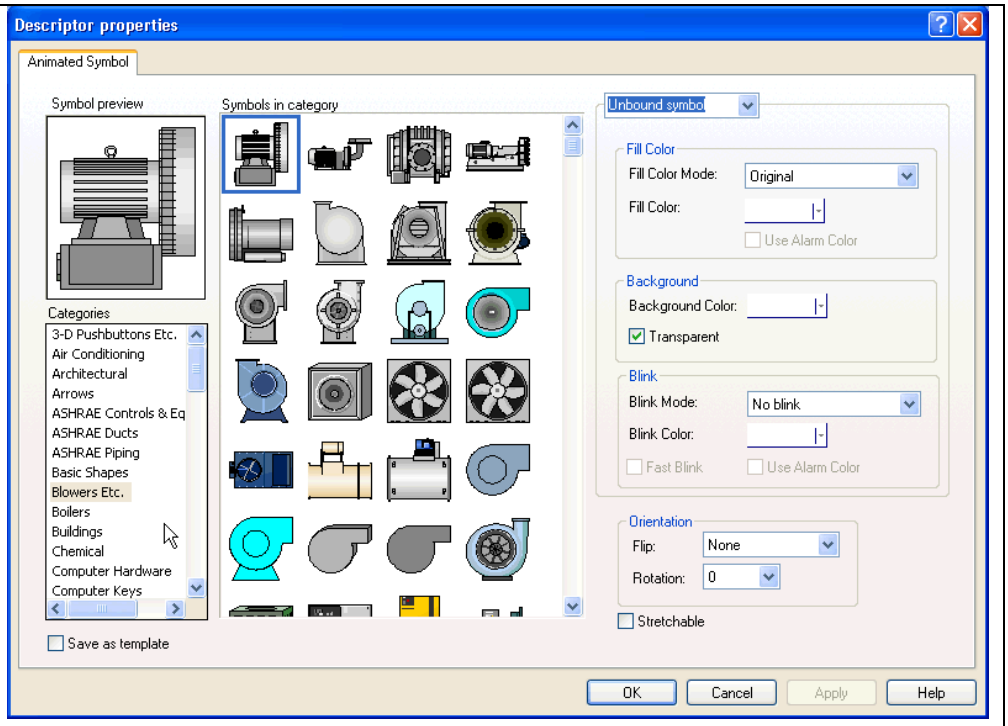
	Data reduction	**Minimum**. and **Maximum**.
	Transfer to History	**Reduced Value**
DISPLAY	**Show**	Under **Name** select **Enable** and under **State/Value** select **Enable**
EDIT MAPPING	**Atom** *Command* *Count* *Preset*	Key in under **PLC addresses for …** DG = 24, WO = 1, BO = 0, Ext. type = FP16S. DG = 23, WO = 0, BO = 0, Ext. type = FP16S. DG = 24, WO = 2, BO = 0, Ext. type = FP16S.
	Alarm Details	Create an alarm text for **Limit:** `Counter limit exceeded`, **Priority** = 10, **Alarm color** red and **Blink** enabled, **Instructions**: `Use auxillary mixer`
ATTRIBUTES OF A RECTANGLE	**Frame and Border**	Select **Enable** and then click on the **Settings** button.
	Frame/Border Settings	Click the button **Frame Color** and choose a dark red color. In **Frame Thickness** choose 5. Click **OK**.to finish and save.

Task 4:
Creating a standard digital pump with Animated Symbol

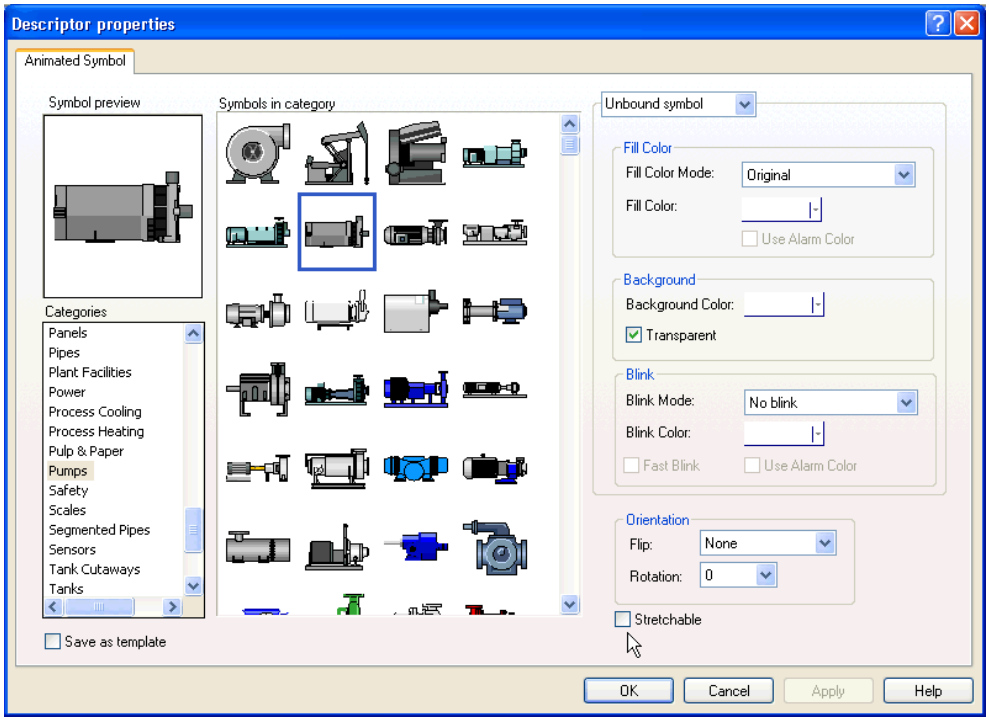
Our next task is to place a simple on-off pump on the mimic and represent it with a graphic from the **Animated Symbol** library.

Step	Action
1.	In the menu go **View → Drawing Toolbar** to open the toolbar.
2.	Find the symbol  and left click on it while dragging it onto the diagram where it appears as a black square.
3.	Click on the black square and a graphic for a blower appears.
4.	Double click the graphic and the properties page for **Animated Symbol** appears.





5. Use the scroll bar on the left to come down to the category **Pumps** and select one of the pumps in the second line.

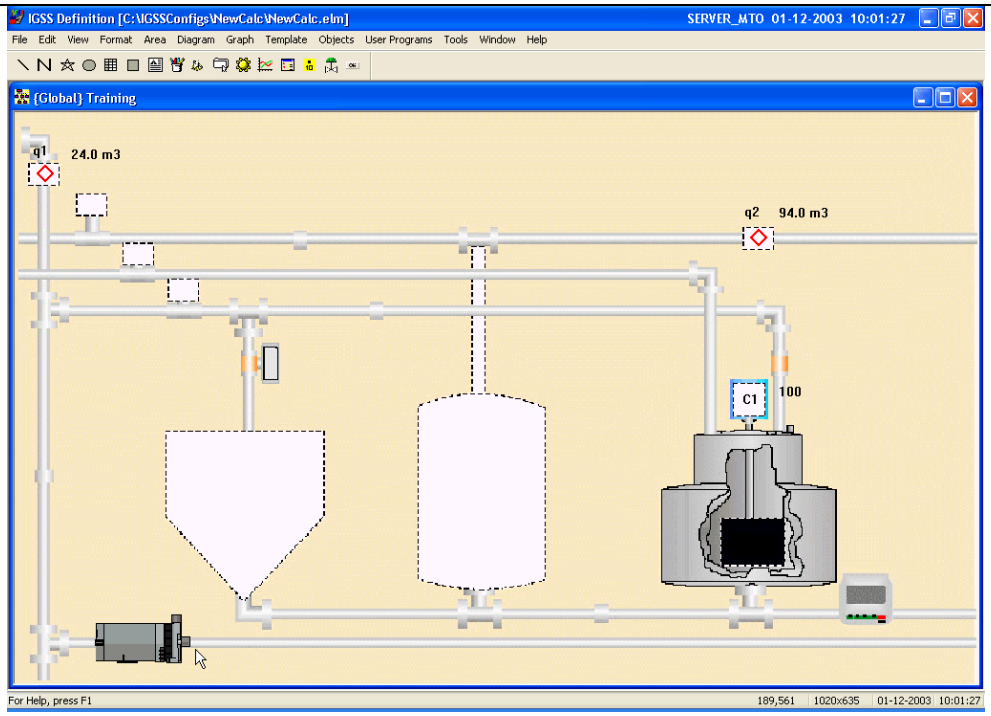


Select the check box **Stretchable** and observe what happens to the graphic.

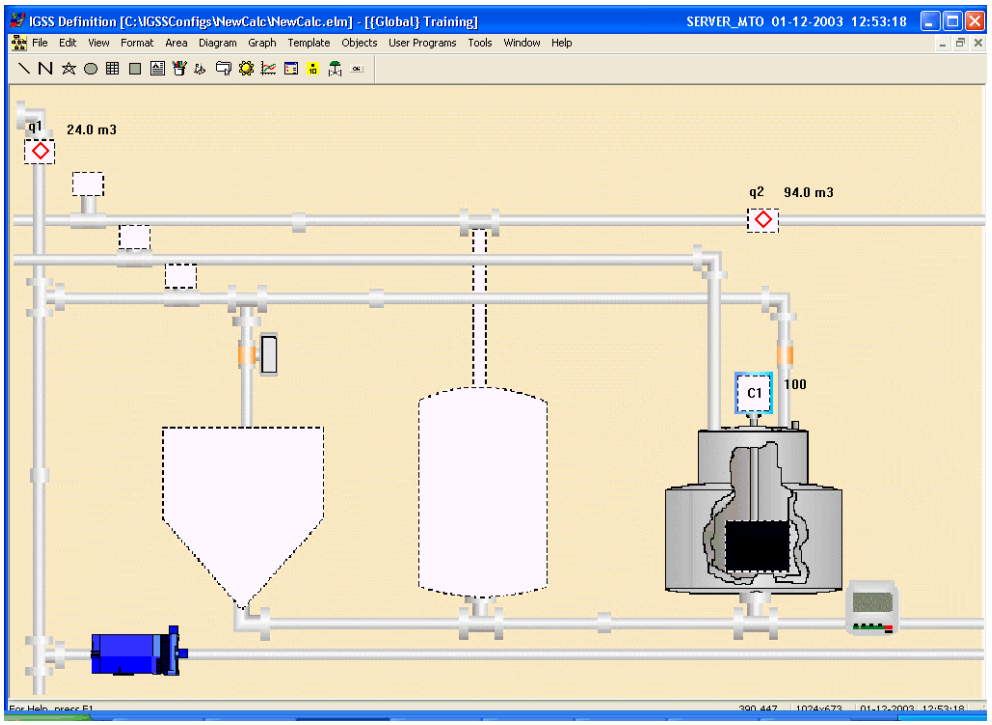
6. Click **OK** and the graphic comes onto the diagram.

7. Position the pump graphic as shown below



	 .
8.	On the menu go **Objects** → **New Unreferenced…** and the **Object Browser** comes up. We are creating a digital object which has no display.
9.	In the tree view on the left make sure you select **Workshop,** click on the + and then click on the folder **Digital** and then **(None)**.
10.	At the bottom of the **Object Browser** under **Name:** type in the name for the pump, `p4` and a description, `2-state`.
11.	Click **Create** and the object properties tabs appear. Notice there's no **Display** tab.
12.	Go to the **Data Management** tab and key in the following: **Scan interval:** 5000 **Base interval:** 5 **Data reduction: Change**
13.	Go on to the **Edit Mapping** tab and choose the following settings. Highlight **Command** and on **I/O mode:** choose **local** Highlight **State** and on **I/O mode:** choose **local**
14.	Click **OK** and the properties tabs close.
15.	On the diagram, right click on the pump graphic and select **Connect…**.



16.	In the **Object Browser** find and select the pump **p4** and click the button **Open/Select** and the pump graphic is connected to the object properties we defined for **p4** and we see now that a new tab has been added to the properties tabs for the pump, namely **Animated Symbol**.
17.	Go to the Animated Symbol tab and do the following: In the upper right hand corner in the drop down box with the default setting **In alarm**, select the **<-0** setting which is the state OFF for **p4**. In **Fill Color Mode** choose **Shaded** and in **Fill Color** choose blue. Return to the drop down box and select the **<-1** setting. **Fill Color Mode: Solid Fill Color:** select dark green and **Background Color:** remove check mark in **Transparent** and select yellow. 
18.	Now go to the **Display** tab and select the following: **Name** select **Enable** **State / Value** select **Enable** **Atom** select **Command**
19.	Click **OK** to finish and save.

Task 5:
Using Calculation on p4

Our final task is to use the **Calculation** function to automatically control the pump, **p4**. Our aim is to ensure that **p4** is stopped every time the flow through the gauge **q1** is greater than a given speed.



Step	Action

1. Click on the **p4** graphic to open the properties tabs and go to the **Calculation** tab..

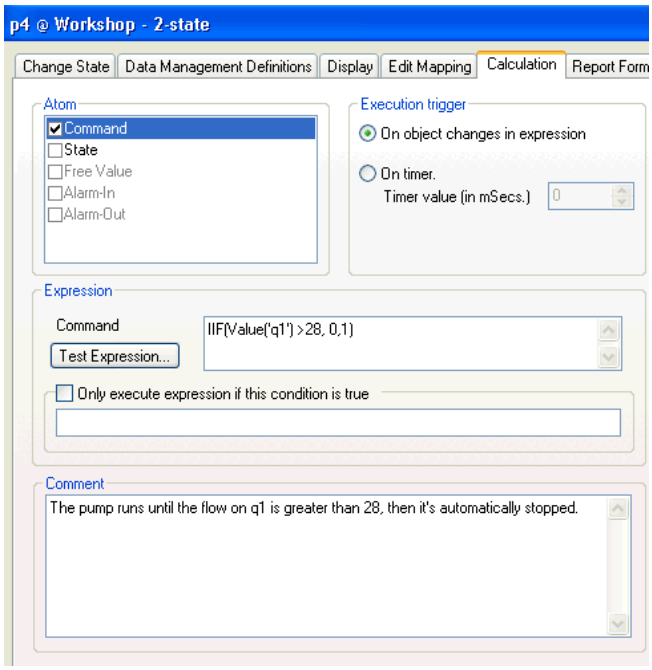
2. Under **Atom** choose **Command** and under the **Execution trigger** group select **On object changes in expression**.

3. In the **Expression** group, key in the following expression in the blank field: `IIF(Value('q1')>28, 0,1)`
(NOTE: There is a space between the comma and the zero in the line above. If not written exactly as shown, an error will be displayed.)
This tells the system that if the value of q1 exceeds 28 m3/hr. then p4 must be stopped (0), otherwise p4 must run (1).

4. Click the button **Test Expression…** to verify that the syntax is correct.

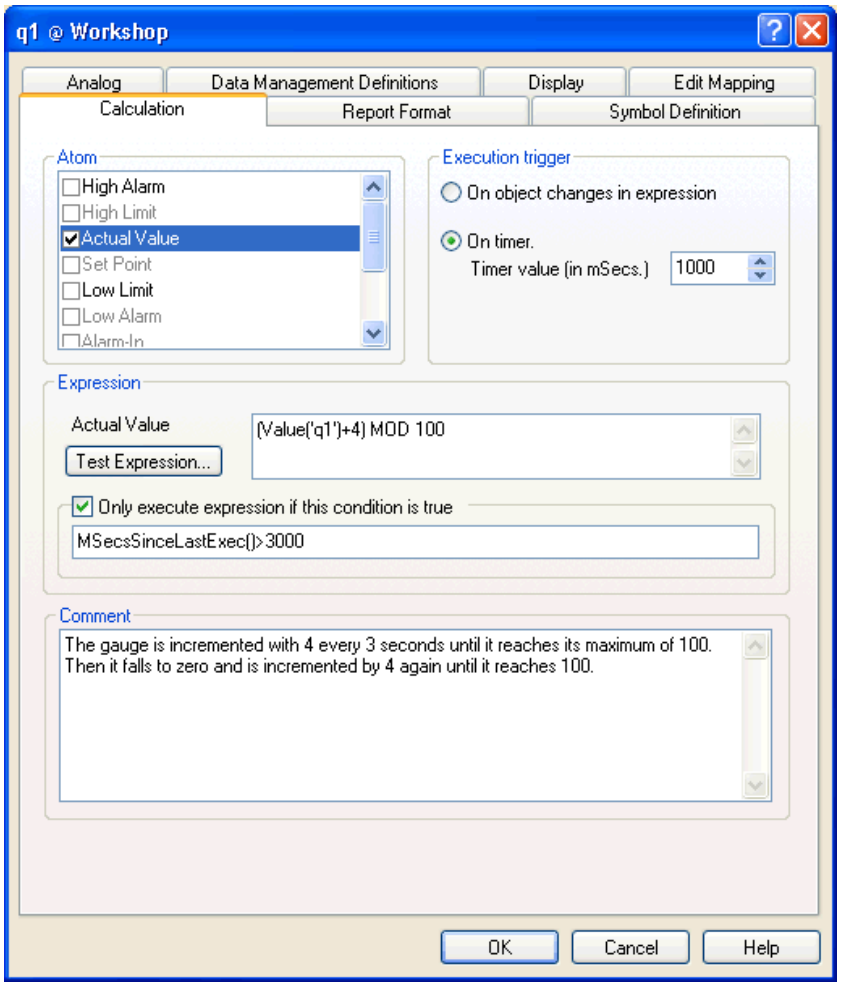
5. In the **Comment** field key in a short description of what's happening, for instance "The pump runs until the flow on q1 is greater than 28, then it's automatically stopped."

6. Click **OK** to complete.





7. Click on the symbol for the flow gauge **q1** and go to the **Calculation** tab and put in the same settings as in the illustration below.



8. When finished with entering the settings, click the **Test Expression…** button to verify correct syntax.

9. Click **OK**.

10. Save and install the configuration and correct any errors.

11. Activate **IGSS Starter** and observe when **p4** is stopped and started again.

EXPLANATION:

Atom we choose the **Actual Value** atom to manipulate with our expression

Execution trigger we choose to have the expression executed on the basis of time intervals, **On timer, Timer value (in mSecs)**in this case 3000 milliseconds



Expression We want the 4 units to be added to the **Actual Value** of the flow gauge **(Value('q1')+4)**, and when the maximum range of **q1** is reached, which is100, then we want to start anew with the **Actual Value** being set to zero, **MOD 100**.

Only execute expression if this condition is true We don't want the addition of 4 units to occur until at least 3000 milliseconds have elapsed since the last time 4 units were added, **MsecsSinceLastExec()>3000**

RESULT: the q1 flow gauge will register an increase in flow with increments of 4 until it reaches 100 at which time it will go back to zero and begin anew.

OPTIONAL TASK:

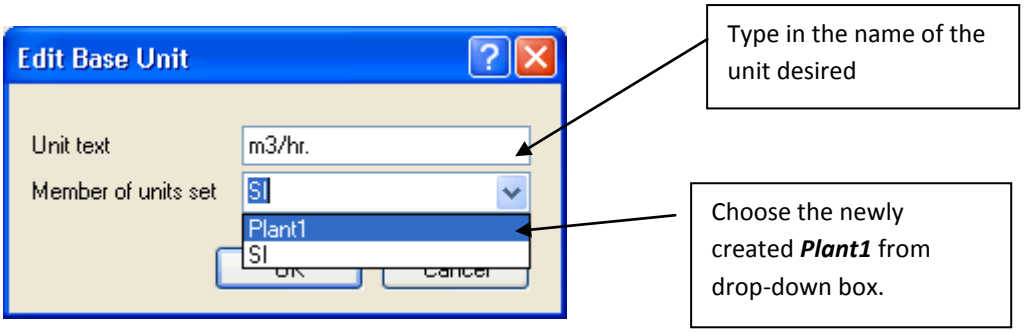
Create a new Units set for use in defining objects

Do this **only** if you have time left after completing the above tasks in this exercise.

We wish to create a set of abbreviations for measurement units, which we will then be able to choose for any object in our project. These will appear on the tab where the name of the object can be changed, for instance on the **Analog** tab for analog objects.

Step	Action
1.	Open **Edit → Unit Sets…**
2.	Click on **New** and key in the name `Plant1` for the new Unit Set**.**
3.	Click **Close** to save and exit.
4.	Next, open **Edit → Measurement Units…**
5.	Click on the button **New** and the **Edit Base Unit** dialogue appears.
6.	In the field **Unit text** key in the name of the measurement unit needed, for instance `m3/hr and in the` **Member of units set** `choose` **Plant1** `in the drop down box.` `Click` **OK.**

The dialog box for creating new units





7.	Repeat step **6.** above to create 2 more unit texts called `hrs.` and `m.`.
8.	Go back to the objects in the **Exercise** configuration and add the appropriate unit in the **Units** field, for instance on **q1** go to the **Analog** tab to find this field and use the drop down box to select the unit.

YOU HAVE SUCCESSFULLY COMPLETED THE EXERCISE !!